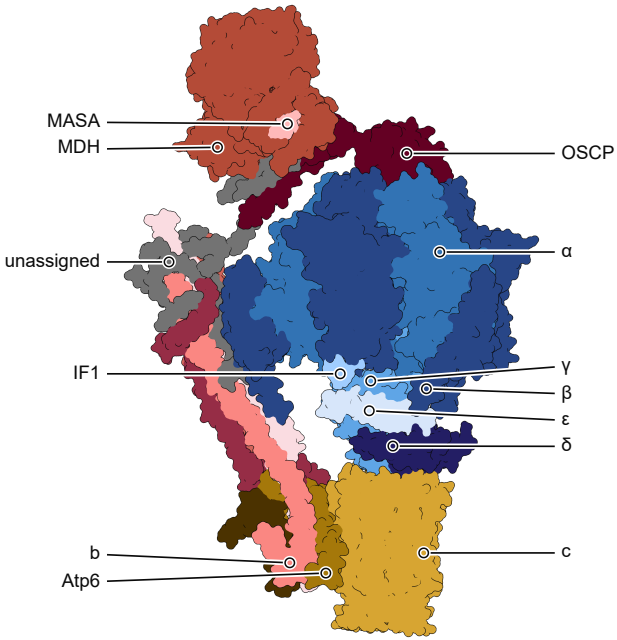
A TCA-cycle enzyme, malate dehydrogenase (MDH), is integrated into Acanthamoeba ATP synthase.

TCA: tricarboxylic acid. Structure PDB 9ZFT, one monomer, coloured by module.

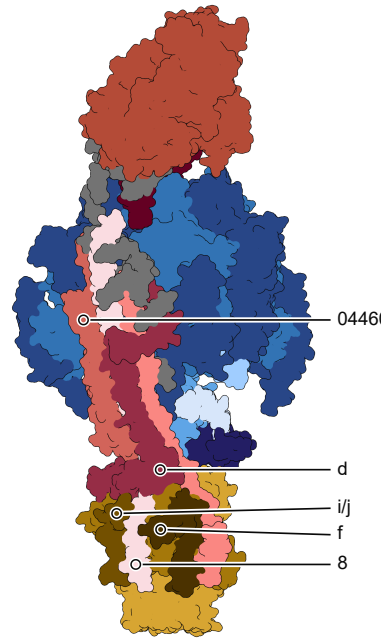
a **Side view of the monomer**

All 34 chains drawn, one surface each.
Rings mark the chain that a label names.



b **Second view of the monomer**

Same colours as a, turned 90° about the vertical axis.



Modules

Family = module, tone = subunit.

F1
catalytic head with γ δ ε, IF1

Fo
membrane part

Stalk
peripheral stalk incl. MASA, MDH

Unassigned
grey, no module

c **Key: subunits by module**

Entity numbers and chain ids are different things; chain ids are case-sensitive.
Panel: where the subunit is labelled, in the view that shows it best. ×N: copies; a label names one.

● identified from map density, * identified by homology

Subunit (× copies)	Panel	Entity	Gene ID	Chain ids (case matters)	Identified
F1 catalytic head					
α ×3	a	16	ACANB15922	a b c	●
β ×3	a	17	ACANB02410	d e f	●
γ	a	3	ACANB07530	A	●
δ	a	5	ACANB02409	C	● †
ε	a	9	ACANB12077	H	*
IF1	a	7	ACANB08262	E	*
Fo membrane part					
Atp6	a	1	ACANB15943	6	●
c ×9	a	14	ACANB15917	P Q R S T U V W X	*
f	b	12	ACANB14194	K	●
i/j	b	11	ACANB04724	J	●

● identified from map density, * identified by homology

Subunit (× copies)	Panel	Entity	Gene ID	Chain ids (case matters)	Identified
Stalk peripheral stalk					
8	b	2	ACANB15938	8	●
b	a	4	ACANB15939	B	●
d	b	6	ACANB10705	D	●
OSCP	a	8	ACANB01783	G	●
04460	b	15	ACANB04460	Z	*
MASA	a	13	ACANB14707	M	●
MDH ×2	a	10	ACANB11954	I L	●
unassigned	a	18 19 20 21	–	1 2 3 5	

Gene ID: as in the entry; Fig. 1F lists ε also as 06500.
04460: no subunit name in the paper; it is a gene number.
MDH: malate dehydrogenase, an enzyme of the tricarboxylic acid cycle.
MASA: MDH-ATP synthase adaptor. IF1: inhibitory factor 1.
OSCP: oligomycin sensitivity-conferral protein.

– : the entry gives no gene ID (entities named unknown protein).
Grey: helices "visible, but unassigned" in the map (paper, Fig. 1D).
† Fig. 1F marks δ de novo, but the Results group δ with c, ε, IF1 and 04460, found by complexomics.
Stalk includes MASA and MDH, the "hat" above OSCP (paper).
Subunits e and g (Fig. 1F) are built in PDB 9ZFW, the membrane part of the dimer, not in 9ZFT.
Example: Atp6 is entity 1, chain id 6; entity 6 is subunit d.

Deposited structure 9ZFT (entry 4.3 Å, paper 4.2 Å); nucleotides, ions, membrane not drawn.

After Fig. 1 of Fry et al., Cell 2026, 189:6307, doi:10.1016/j.cell.2026.08.057, CC BY 4.0. Data: PDB 9ZFT. Redesigned; values as published.